Project Planning & Work Breakdown Structure

Date	15 March 2026
Team ID	Rahul-DS633
Project Name	CrimeVision AI - Multimodal Crime Scene Reconstruction
Maximum Marks	5 Marks

1. Project Milestones & Timeline (12-Week Roadmap)

Milestone	Phase / Activity	Duration	Key Deliverables	Status
M1	Problem Definition & Requirements Analysis	Weeks 1 - 2	Problem statements, customer journey map, DFD, technology stack specification.	**Completed**
M2	Computer Vision & 3D Spatial Pipeline	Weeks 3 - 4	RT-DETR object detection, depth map estimation, Three.js 3D canvas viewport.	**Completed**
M3	Audio Intelligence & Speech Transcription	Weeks 5 - 6	Whisper ASR integration, acoustic event detection, 4-language translation tabs.	**Completed**
M4	CCTV Video Temporal Timeline Engine	Weeks 7 - 8	1.0 Hz video frame decomposition, BLIP captioning, frame-by-frame timeline.	**Completed**
M5	Multimodal Fusion & Groq Reasoning	Weeks 9 - 10	Cross-modal timestamp synchronization, Groq LLaMA-3.3 forensic chatbot.	**Completed**
M6	Court PDF Reporting, Testing & Release	Weeks 11 - 12	ReportLab PDF generator, unit test suite (5/5 passing), external data validation.	**Completed**

2. Risk Management Matrix

Risk Description	Severity	Likelihood	Mitigation Strategy Implemented
Large deep learning models failing on low-end hardware.	High	Medium	Implemented dual-tier lazy loading with local cache fallback and lightweight CV heuristic detectors.
Audio transcripts containing unescaped characters breaking PDF generation.	High	High	Applied comprehensive xml.sax.saxutils.escape sanitation across all ReportLab Paragraph injections.
Browser Three.js canvas crash when switching hidden tabs.	Medium	Medium	Added zero-dimension guards inside onResize() to prevent NaN camera aspect ratios.